

Mayank Pandey

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Professional Summary:

With over 2.2 years of experience as a **Member Technical Staff**, I possess specialized expertise in **Python, Machine learning, Generative AI, Large Language Models (LLMs), Deep Learning, NLP, Prompt Engineering**, and **Backend Development**. My work focuses on driving innovation through cutting-edge research and experimentation with advanced NLP techniques and machine learning algorithms. I thrive in collaborative environments, actively contributing to knowledge-sharing and driving continuous learning.

Work Experience:

Project Name: iSee

Client: NEC Japan

Experience: 2.2 years

Company: NEC India Corporation Limited

Current Role: Member of Technical Staff

Industries: SAP, Backend Development, GenAI

Key Achievements in the Project (iSee):

- **Research & Development:** Continuously engage in research and experimentation with **NLP techniques, including transformer models, pre-trained language models, and Generative AI**.
- **NLP Expertise:** Applied NLP techniques to solve complex business challenges, including named entity recognition and text classification, with a focus on natural language understanding.
- **Backend development in python:** Extensive experience in backend development using Python, including designing and implementing APIs, database management, and system optimization.
- **Fine-tuning & Model Evaluation:** Led the fine-tuning and evaluation of **Large Language Models (LLMs)** for tasks such as keyword/keyphrase extraction, ensuring optimal performance and accuracy.
- **Model Retraining Implementation for Enhanced Prediction Accuracy:** Implemented **model retraining** in both **UI & backend**, enabling continuous improvement by incorporating new data, Achieved **40% higher** prediction accuracy.
- **Deployment of Nous-Hermes-Lama Model in iSee Project:** Deployed the **Nous-Hermes-Llama** model in **iSee**, enhancing **NLU** and **response generation**, leading to **more accurate, context-aware insights**.
- **Role-based RAG System:** Developed and worked on an advanced **Role-based Retrieval-Augmented Generation (RAG)** system, allowing users to query from LLMs and receive responses based on a vector database.
- **Prompt Engineering:** I developed an **Automatic Prompt Generation Approach** that selects examples and evaluates them using custom auto-prompt templates. To enhance prompt efficacy, I implemented advanced techniques like **Zero-Shot, One-Shot, and Two-Shot Learning** for adaptable query handling, **Chain of Thought** for better multi-step decision-making, **Tree of Thoughts** for structured reasoning, and **Dynamically Stimulated Prompts** that adapt to user input for more responses.
- **Model Design:** Designed and implemented a model using **TensorFlow** and **PyTorch** to convert business text into SAP.
- **Project Contributions:** In the **iSee Project**, I developed new features to enhance **functionality & user experience**, fixed **legacy bugs** for better **stability & performance**, and Optimized recommendations with **cosine similarity** for **data matching**.

Skills:

- **Programming Languages:** Python
- **Machine Learning:** Supervised & Unsupervised Learning, Classification Algorithms (e.g., Logistic Regression, SVM, Random Forest, XGBoost), Transfer Learning, Fine-Tuning
- **Natural Language Processing (NLP):** Named Entity Recognition (NER), Text Classification, Keyword Extraction
- **AI & Deep Learning:** Generative AI, Transformers, LLMs, TensorFlow, PyTorch, GPT-3.5, Nous-hermes-lama2, Solar (SOLAR - A LLM from Hugging Face), Mistral (Mistral AI's open-weight models), Named Entity Recognition (NER)
- **Libraries/Frameworks:** Langchain, LlamaIndex, Scikit-learn, Keras, Flask, FastAPI
- **Data Modeling & Database:** MySQL, SQLAlchemy, SQLite
- **Data Visualization:** Matplotlib, Seaborn, Pyplot
- **Web Scraping & Automation:** Selenium, BeautifulSoup, Linux command-line scripting
- **Tools & Technologies:** Retrieval-Augmented Generation (RAG), Cosine Similarity, Prompt Engineering
- **APIs:** REST API, Flask, Falcon, django
- **Data Analysis:** Power bi Desktop

Education: Bachelor Of Technology in ECE

Passing Year :2023

Projects:

- **Custom AI Chatbot for E-commerce** - Developed a GPT-4 chatbot with **LangChain** and **RAG** for customer support and product recommendations.
- **Sales Analysis Dashboard for E-Commerce (Power BI)** - Developed a Power BI dashboard to analyse sales trends, revenue, and customer behavior using SQL, Excel, and DAX.